DDoS ATTACK DETECTION

USING LSTM + AUTOENCODER-BASED ANOMALY DETECTION

A deep-learning approach for modeling network-flow sequences and detecting anomalous traffic patterns.

PROJECT DETAILS

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SECTION / BRANCH	Section 4 • CSE
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ABSTRACT

Project motivation and proposed learning strategy

Distributed Denial-of-Service (DDoS) attacks overwhelm network services with malicious traffic and remain a major threat to availability, reliability, and cyber security. This project proposes a hybrid deep-learning framework that combines Long Short-Term Memory (LSTM) networks with an Autoencoder-based anomaly detector. The LSTM component models temporal and sequential patterns in network-flow data, while the Autoencoder learns a compact representation of normal behaviour and flags deviations using reconstruction error.

The project report identifies CICDDoS2019 as the intended benchmark and evaluates the system using Accuracy, Precision, Recall, F1-score and ROC-AUC. The objective is to detect both known and previously unseen attack variations without depending only on predefined signatures.

TEMPORAL LEARNING · ANOMALY DETECTION · ZERO-DAY FOCUS · NETWORK SECURITY

INTRODUCTION

Why a hybrid LSTM–Autoencoder architecture is relevant

WHY DDoS DETECTION MATTERS

The rapid growth of internet-connected devices and cloud services has increased the attack surface for DDoS campaigns. Successful attacks can make critical services unavailable, interrupt business operations, and cause financial and reputational damage. Static signature-based defenses often struggle when attack behaviour changes or previously unseen patterns appear.

WHY LSTM + AUTOENCODER

LSTM networks are designed to learn sequential dependencies in time-ordered data, making them suitable for network traffic. An Autoencoder can learn normal traffic behaviour without requiring attack signatures; unusually high reconstruction error can indicate anomalous flows. Combining these ideas creates a detection pipeline that can model temporal patterns and also identify deviations from learned normal behaviour.

TRAFFIC → SCALE → SEQUENCE → LSTM → AUTOENCODER → ANOMALY

REQUIREMENT ANALYSIS

Core tools needed to execute the proposed pipeline

PYTHON 3.x	Programming language for data preparation and model development
TENSORFLOW / KERAS	Deep-learning framework for LSTM and Autoencoder architectures
PANDAS • NUMPY • SCIKIT-LEARN	Data handling, preprocessing, scaling and evaluation
JUPYTER / PYTHON IDE	Interactive development and experimentation environment
GPU RECOMMENDED	Reduces training time for deep-learning experiments

LITERATURE REVIEW • PART 1

Papers 1–5

PAPER / YEAR	DATASET	METHODOLOGY	LIMITATION
1. Reconstruction-based LSTM-Autoencoder for Anomaly-based DDoS Attack Detection over Multivariate Time-Series Data Y. Wei, J. Jang-Jaccard, F. Sabrina, A. Singh, W. Xu, S. Camtepe • 2023	CICDDoS2019 (reflection-based: DNS, LDAP, SNMP)	LSTM-Autoencoder (LSTM-AE) with reconstruction-error thresholding for unsupervised anomaly detection	Evaluated only on reflection-based attack types; limited coverage of the full CICDDoS2019 attack diversity
2. DDoSNet: A Deep-Learning Model for Detecting Network Attacks M. S. Elsayed, N.-A. Le-Khac, S. Dev, A. D. Jurcut • 2020	CICDDoS2019	Recurrent Neural Network (RNN) combined with an Autoencoder and Softmax classifier	Designed and validated only for SDN environments; generalization to traditional networks not tested
3. Efficient Detection of DDoS Attacks Using a Hybrid Deep Learning Model with Improved Feature Selection D. Alghazzawi, O. Bamasag, H. Ullah, M. Z. Asghar • 2021	CICDDoS2019	Hybrid CNN + Bidirectional LSTM (BiLSTM) with Chi-square-based feature selection	Accuracy (94.52%) is lower than several later hybrid/LSTM-based models; performance highly dependent on the feature-selection step
4. Evaluating the Performance of LSTM and GRU in Detection of Distributed Denial of Service Attacks Using CICDDoS2019 Dataset M. Subrmanian, K. Shanmugavadivel, P. S. Nandhini, R. Sowmya • 2022	CICDDoS2019	Comparative supervised classification using LSTM and GRU networks	Purely supervised approach with no anomaly-detection component for unseen attacks; GRU accuracy (92.5%) notably lower than LSTM (99.4%)
5. Improved DDoS Detection Utilizing Deep Neural Networks and Feedforward Neural Networks as Autoencoder A. L. Yaser, H. M. Mousa, M. Hussein • 2022	NSL-KDD and CICDDoS2019	Feedforward Neural Network stacked as an Autoencoder combined with a Deep Neural Network classifier	Feedforward autoencoder cannot model temporal/sequential dependencies present in network traffic flows

LITERATURE REVIEW • PART 2

Papers 6–10

PAPER / YEAR	DATASET	METHODOLOGY	LIMITATION
6. A Long Short-Term Memory (LSTM)-Based Distributed Denial of Service (DDoS) Detection and Defense System Design in Public Cloud Network Environment (LSTM-CLOUD) R. Aydin • 2022	CICDDoS2019	LSTM-based signature detection module paired with a defense/mitigation module	Signature-based detection strategy limits the system's ability to identify novel or zero-day attack patterns
7. An Improved Deep Learning Model for DDoS Detection Based on Hybrid Stacked Autoencoder and Checkpoint Network A. K. Mousa, M. N. Abdullah • 2023	SDN DDoS dataset and Botnet dataset	Hybrid LSTM + CNN with a stacked autoencoder, combined with a checkpoint network for fault tolerance	Increased architectural complexity and training overhead from stacking multiple deep components; not evaluated on CICDDoS2019 directly
8. Improving Performance of Autoencoder-Based Network Anomaly Detection on NSL-KDD Dataset W. Xu, J. Jang-Jaccard, A. Singh, Y. Wei, F. Sabrina • 2021	NSL-KDD	Deep autoencoder with feature reduction for unsupervised network anomaly/intrusion detection	Tested only on the older NSL-KDD dataset; does not address modern, high-volume DDoS traffic patterns found in CICDDoS2019
9. Hybrid Deep Learning Approach for Automatic DoS/DDoS Attacks Detection in Software-Defined Networks H. Elubeyd, D. Yiltas-Kaplan • 2023	CICDDoS2019 and a custom-generated SDN dataset	Hybrid combination of 1D CNN, Gated Recurrent Unit (GRU), and Dense Neural Network (DNN)	Combining three deep learning components increases training/computational cost; accuracy varies noticeably between the two datasets used (99.81% vs 99.88%)
10. An Efficient Hybrid-DNN for DDoS Detection and Classification in Software-Defined IIoT Networks A. Zainudin, L. A. C. Ahakonye, R. Akter, D.-S. Kim, J.-M. Lee • 2023	Software-Defined IIoT network traffic dataset	Hybrid CNN-LSTM model with XGBoost-based feature selection	Purpose-built for Software-Defined IIoT environments; feature-selection dependency may limit robustness to unseen/evolving attack types

LITERATURE REVIEW • PART 3

Papers 11–15

PAPER / YEAR	DATASET	METHODOLOGY	LIMITATION
11. MSCNN-LSTM-AE: Hybrid Autoencoder Combining Multi-Scale CNN and LSTM for Network Anomaly Detection T. Singha, J. Jang-Jaccard • 2022	UNSW-NB15, NSL-KDD, and CICDDoS2019	Two-stage autoencoder: Multi-Scale CNN Autoencoder (MSCNN-AE) for spatial features followed by an LSTM-Autoencoder for temporal features	Reported accuracy (93.76%) and recall (92.26%) are lower than several supervised alternatives; two-stage pipeline adds architectural complexity
12. Deep Learning for Cyber Security Intrusion Detection: Approaches, Datasets, and Comparative Study M. A. Ferrag, L. Maglaras, S. Moschoyiannis, H. Janicke • 2020	Multiple benchmark datasets (CICIDS2017, NSL-KDD, and others)	Comparative survey and benchmarking of deep learning architectures (including LSTM and Autoencoders) for intrusion detection	A survey/benchmark study rather than a novel detection framework; does not propose a combined LSTM-Autoencoder architecture evaluated on CICDDoS2019
13. LSTM-Based Collaborative Source-Side DDoS Attack Detection S. Yeom, C. Choi, K. Kim • 2022	Simulated/collected source-side network traffic (IEEE Access study)	LSTM-based classifier deployed collaboratively across multiple source-side detection points	Requires deployment and coordination across multiple source nodes, increasing system complexity; not evaluated on CICDDoS2019 or with autoencoder-based anomaly detection
14. Detection of Known and Unknown DDoS Attacks Using Artificial Neural Networks A. Saied, R. E. Overill, T. Radzik • 2016	Custom-generated DDoS traffic dataset	Feedforward Artificial Neural Network (ANN) trained on labelled traffic signatures	Shallow ANN lacks temporal/sequential modeling and unsupervised anomaly detection, limiting effectiveness against fully unknown (zero-day) attacks
15. LUCID: A Practical, Lightweight Deep Learning Solution for DDoS Attack Detection R. Doriguzzi-Corin, S. Millar, S. Scott-Hayward, J. Martinez-del-Rincon, D. Siracusa • 2020	CICDDoS2019, ISCX2012, and other benchmark datasets	Lightweight Convolutional Neural Network (CNN) with dataset-agnostic preprocessing for low-overhead traffic classification	Purely supervised CNN-based classification; no unsupervised/anomaly-detection mechanism, so it struggles to generalize to entirely unseen attack types

LITERATURE REVIEW • PART 4

Papers 16–20

PAPER / YEAR	DATASET	METHODOLOGY	LIMITATION
16. Chronos: DDoS Attack Detection Using Time-Based Autoencoder M. A. Salahuddin, V. Pourahmadi, H. A. Alameddine, M. F. Bari, R. Boutaba • 2021	Network traffic flow datasets (time-windowed)	Time-based Autoencoder using reconstruction error over sliding time windows	Does not incorporate LSTM-based sequence modeling; detection accuracy is sensitive to the choice of time-window size
17. A Deep Learning GRU-BiLSTM for DDoS Attack Detection A. M. Al-Eryani, F. A. Omara, E. Hossny • 2025	CIC-DDoS2019	Hybrid Gated Recurrent Unit (GRU) and Bidirectional LSTM (BiLSTM) model for combined spatial-temporal feature extraction	High computational complexity from combining two recurrent architectures; not evaluated against fully unsupervised/zero-day attack scenarios
18. SDN and Application Layer DDoS Attacks Detection in IoT Devices by Attention-Based Bi-LSTM-CNN I. Priyadarshini, P. Mohanty, A. Alkhayyat, R. Sharma, S. Kumar • 2023	IoT/SDN application-layer traffic dataset	Attention-based hybrid Bi-LSTM and CNN model for application-layer DDoS detection	Focused specifically on application-layer attacks in IoT/SDN settings; attention mechanism adds computational overhead, limiting real-time deployment
19. Multi-Class Intrusion Detection System in SDN Based on Hybrid BiLSTM Model M. Cui et al. • 2024	SDN network traffic dataset (CICIDS2017/CICDDoS2019-based)	Hybrid Bidirectional LSTM (BiLSTM) model for multi-class DDoS/intrusion classification	Multi-class classification increases model complexity and computational cost; performance degrades for underrepresented attack classes
20. A New DDoS Attack Detection Model Based on Improved Stacked Autoencoder and Gated Recurrent Unit for Software Defined Network H. Wang et al. • 2025	Software-Defined Network (SDN) traffic dataset	Improved stacked Autoencoder combined with a Gated Recurrent Unit (GRU) classifier	Added architectural complexity from stacking autoencoder and GRU layers; recent publication with limited independent cross-dataset validation

EDA • DATASET QUALITY

Exploratory analysis of the supplied network-flow CSV

225,745	85	2	4	2
RAW RECORDS	FEATURE COLUMNS	DUPLICATES	MISSING CELLS	CLASSES

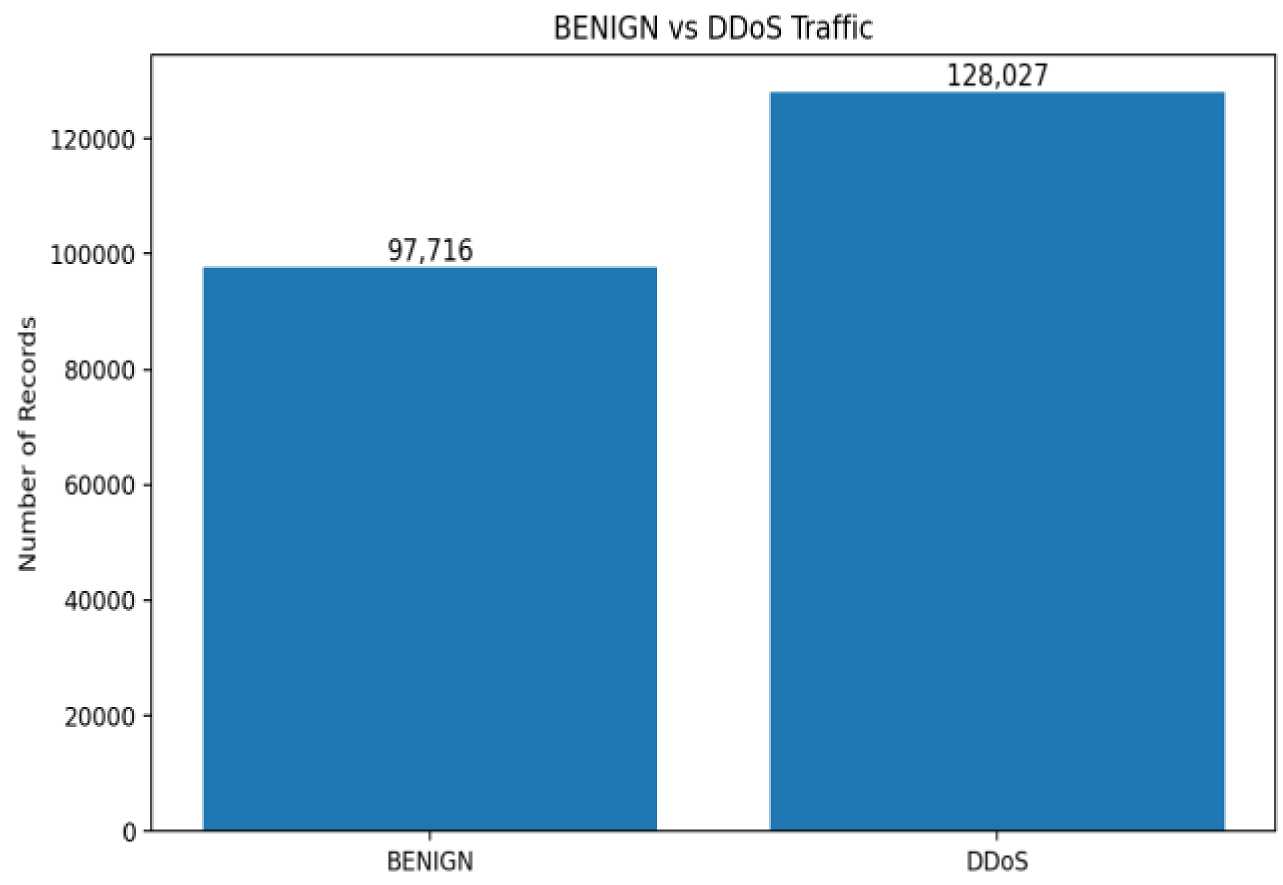
EDA PROCESS

- Inspect dimensions, feature names and data types
- Measure missing values and duplicate network flows
- Remove duplicate rows and preserve the target label
- Compare BENIGN and DDoS class distribution
- Analyse representative numerical traffic features
- Visualize distributions, boxplots, class means and correlation

Only Flow Bytes/s contains missing values in the supplied raw file; the target label has no missing values.

EDA • CLASS DISTRIBUTION

Traffic balance after duplicate removal



CLEANED DATASET

225,743

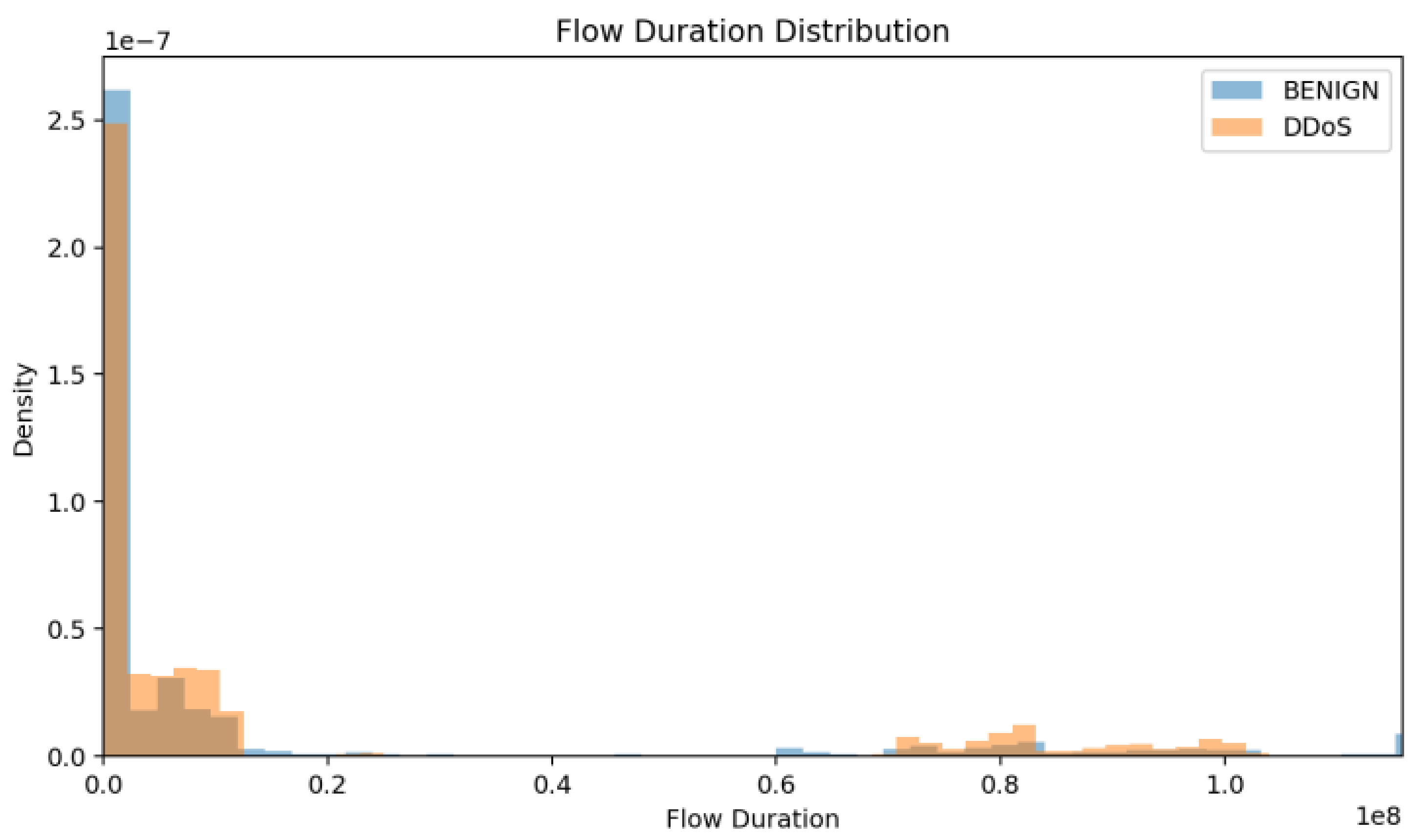
network-flow records

DDoS 128,027 • 56.71%
BENIGN 97,716 • 43.29%

The two classes are reasonably represented, but the DDoS class is larger. Evaluation should therefore use more than accuracy alone.

EDA • FLOW DURATION

Comparing temporal behaviour of BENIGN and DDoS traffic

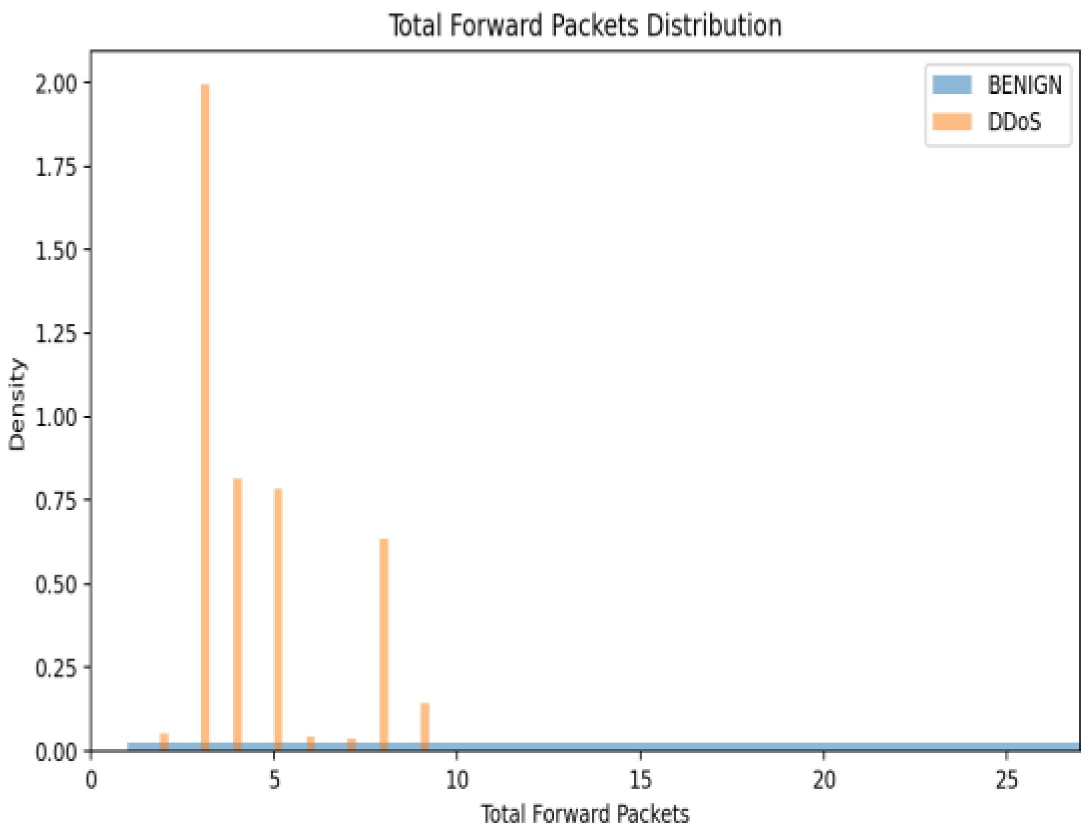


WHAT THIS SHOWS

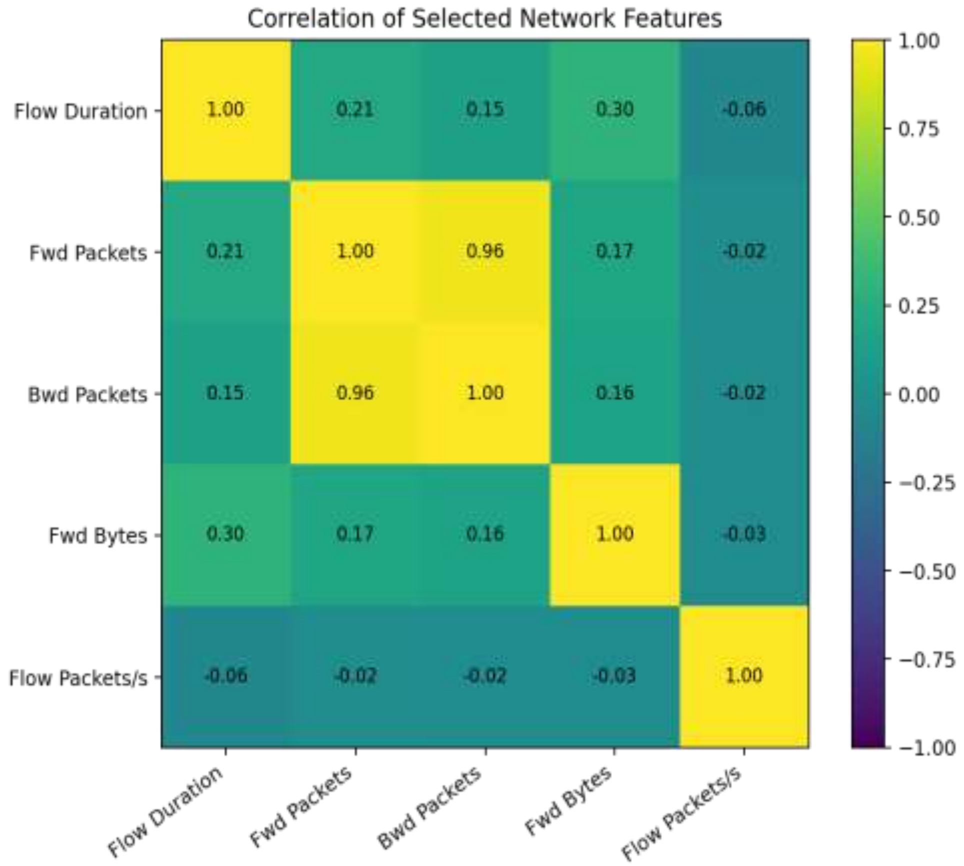
- Flow Duration measures how long a network flow remains active.
- The notebook limits the x-axis to the 99th percentile so extreme values do not dominate the visualization.
- Differences in distribution indicate that temporal flow behaviour contains useful information for distinguishing normal and attack traffic.

EDA • TRAFFIC FEATURES

Packet-volume patterns and feature relationships



TOTAL FORWARD PACKETS



CORRELATION HEATMAP

DATASET DESCRIPTION

What is actually present in the supplied project data

SUPPLIED EDA FILE

Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv

RAW SHAPE	225,745 rows × 85 columns
TARGET	Label → BENIGN / DDoS
CLEANED ROWS	225,743 after removing 2 duplicates
MISSING DATA	4 missing cells in Flow Bytes/s
FEATURE TYPES	Flow duration, packet counts, bytes, rates, flags, windows, active/idle statistics

SOURCE NOTE

The project abstract identifies CICDDoS2019 as the intended benchmark dataset. The EDA file supplied with this presentation is the CSV named above. This slide keeps both source statements separate rather than treating them as the same dataset.

BENIGN vs DDoS

PROPOSED MODEL WORKFLOW

Hybrid temporal learning + reconstruction-based anomaly detection

1	2	3	4	5	6
NETWORK FLOWS	PREPROCESS	SEQUENCE	LSTM	AUTOENCODER	DECISION
Load labelled traffic records	Clean, scale and normalize	Create ordered traffic windows	Learn temporal dependencies	Reconstruct normal behaviour	Use reconstruction error + classifier output

Evaluation: Accuracy • Precision • Recall • F1-score • ROC-AUC

CONCLUSION

Key takeaways from the project foundation

This project establishes a deep-learning framework for DDoS detection by combining the temporal modeling ability of LSTM networks with reconstruction-based anomaly detection from an Autoencoder. The supplied EDA confirms that the network-flow data contain clear class structure and rich numerical traffic features suitable for sequential modeling and anomaly analysis.

The literature review shows that LSTM, CNN, GRU and Autoencoder-based methods are widely used for modern intrusion detection, while many studies still face limitations in zero-day generalization, computational cost and cross-dataset validation. The proposed hybrid direction therefore focuses on both learned temporal behaviour and deviations from normal traffic.

END OF PRESENTATION